

Estimation of Serum Cholesterol

Learning Objectives

By the end of demonstration, students will be able to:

a) Explain the biomedical importance of lipid profile
b) Define hypercholesterolemia with its causes
c) Understand the principle of estimation of blood cholesterol

LIPID PROFILE

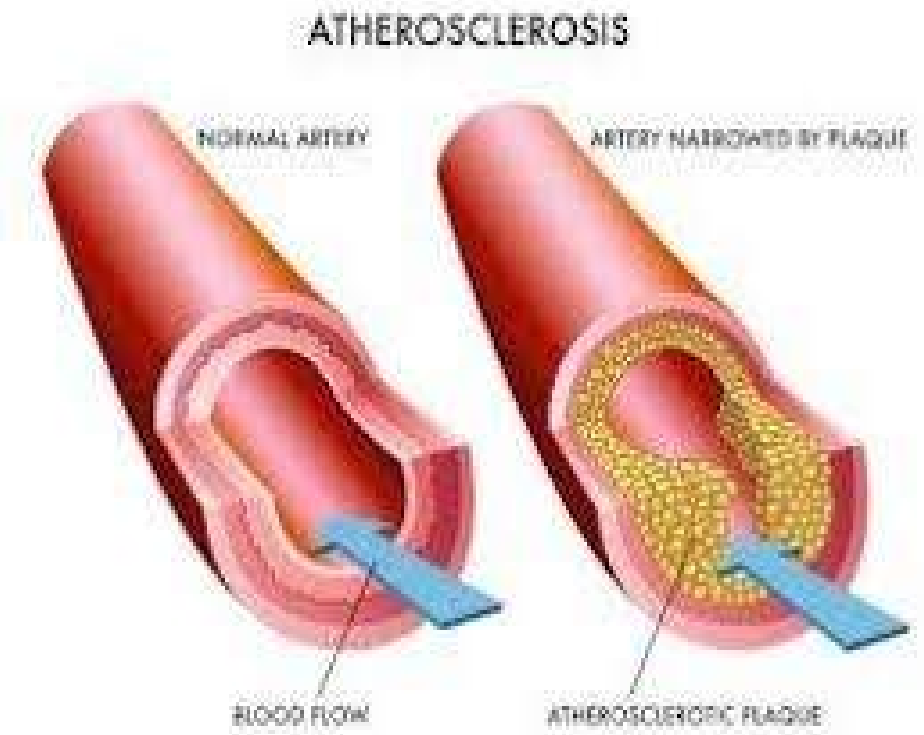
	DESIRABLE	BORDERLINE	HIGH RISK
Cholesterol	<200 mg/dl	200-239 mg/dl	240 mg/dl
Triglycerides	<150 mg/dl	150-199 mg/dl	200-499 mg/dl
HDL cholesterol	60 mg/dl	35-45 mg/dl	<35 mg/dl
LDL cholesterol	60-130 mg/dl	130-159 mg/dl	160-189 mg/dl
Cholesterol/ HDL ratio	4.0	5.0	6.0

- **Lipid profile** or *lipid panel*, is a panel of blood tests that serves as an initial broad medical screening tool for abnormalities in lipids, such as cholesterol and triglycerides

- This test is used to identify hyperlipidemia, many forms of which are recognized risk factors for cardiovascular disease.

- Lipoproteins include:

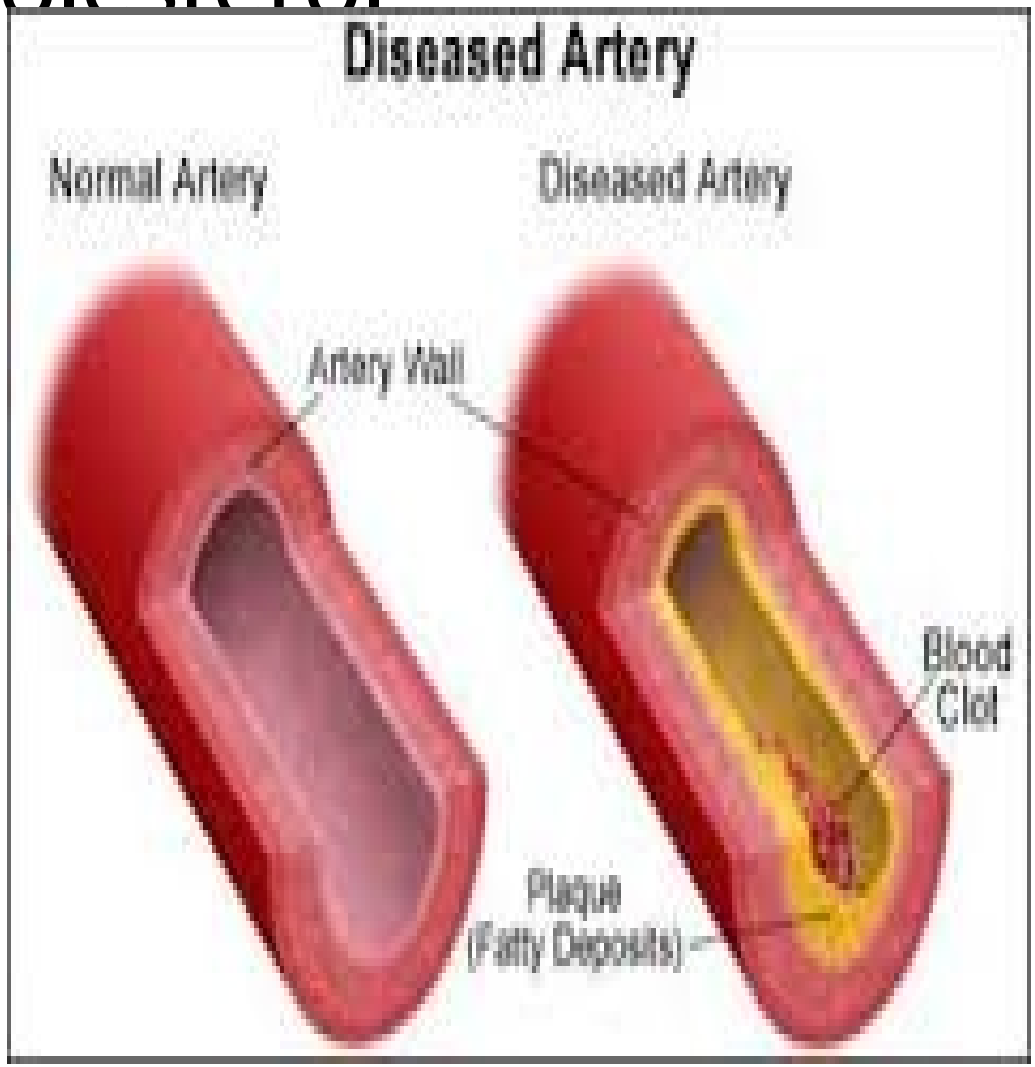
a) Chylomicrons
b) VLDL
c) LDL
d) HDL




Coronary risk factors
high cholesterol
smoking
older age
family history
male gender
diabetes
high blood pressure

Risks associated with raised Cholesterol

- Coronary artery disease (CAD)
- Cerebrovascular accident (CVA)
- Pancreatitis



Risk factors for CAD

- Low HDL Cholesterol (<40mg/dl)
- Hypertension
- Family history
- Diabetes Mellitus
- Cigarette smoking
- Age (male 40 years older)
  (female 55 years older)

Increased cholesterol	Decreased cholesterol
Sedentary lifestyle	Increased exercise
Obesity	High fiber diet
Smoking	Diet low in simple sugars and fried food
Alcohol intake	Omega-3-fatty acids from fish, seed oils
Type 2 DM	
Genetic Factors	



Thank You